

Retrieval & RAG

Retrieval-Augmented Generation, From Parts — Build week

What this week is about

Out of the box, an AI chatbot answers from what it learned on the public internet. It knows nothing about your company's documents — and if you ask, it will confidently make something up. This week you'll fix that: feed the AI your own documents so it answers with real facts and shows where each answer came from.

Why it matters

Why can't the AI just answer from what it already knows?

It was trained on public data, not your files. Ask it about your refund policy and it will invent one.

What is “RAG”?

Retrieval-Augmented Generation — a fancy name for a simple idea: let the AI read the right documents first, then answer from them.

Where would I actually use this?

Support bots, internal search, “ask our handbook” tools — anywhere the answer must come from specific documents.

What you'll learn

- Loading your own documents (PDFs, web pages) into the app
- Splitting documents into small pieces called “chunks” — and why the size matters
- What an “embedding” is: turning each chunk into numbers so the app can match meaning, not just keywords
- Storing those chunks so the app can search them quickly
- Finding the right chunk to answer a question, and checking it before the AI writes anything
- Making the AI quote its source, and say “I don't know” instead of guessing

Topics covered — the exact concepts to study

Why RAG	Embeddings & dense retrieval
Bi-encoder vs cross-encoder	Embedding models (MTEB, BGE, E5)
Chunking strategies	Chunk size & overlap

Vector databases (HNSW)	Qdrant / Chroma / pgvector
Similarity search & top-K	Metadata filtering
Grounded generation & citations	

Your task this week

Evaluated Week 4 · Monday

Build a mini “ask my documents” app. Load a set of real documents, let a user ask a question, and answer using only those documents — with a link to the source. If the answer isn't in the documents, it should say so instead of inventing one.

Same task, six different topics — so everyone works on their own version. Download whichever you're assigned.

- A Customer support tickets**
Ingest the new help-centre drop and prove your chunking finds the answer

- B Recipes & food**
Ingest the new cookbook chapter and prove your chunking finds the answer

- C HR policy**
Ingest the new policy addenda and prove your chunking finds the answer

- D Insurance claims**
Ingest the new endorsement pack and prove your chunking finds the answer

- E Developer documentation**
Ingest the new SDK reference pages and prove your chunking finds the answer

- F Legal contracts**
Ingest the new amendment pack and prove your chunking finds the answer

How your mentor checks

Friday · no marks

A quick review of your task. Not a test — just a few things your mentor looks at, so you can fix anything before Monday.

- Can the app answer a question correctly using the documents?
- Does every answer show which document it came from?
- Ask it something not in the documents — does it admit it doesn't know?

- Did they try more than one chunk size and notice the difference?